

Spirit Scheduling Engine

Comprehensive Development Status & Architecture Report — Phase 1, 2, & 3.1 Complete

Date: August 25, 2026	Target OS/Env: macOS / Python 3.12	Branch: spirit-scheduling-development
Test Suite: 66 / 66 Passed (100%)	Code Quality: 0 Ruff / Django Errors	Secret Check: Passed (0 Leaks)

1. Executive Summary

The **Spirit Scheduling Engine** is a custom Django management application designed specifically for **Spirit of Newfoundland Productions**. It automates complex show staffing by matching show dates, guest counts, staff availability, role capabilities, weekend rotation rules, and fairness history to generate deterministic, audit-backed roster recommendations.

Development is **100% complete through Phase 3.1**. The application features a Bootstrap 5 web portal, public calendar show scraping, CSV availability importing, weekend office & 50/50 rotation engines, audited management overrides, local approval workflows, multi-format exports (Excel, PDF, CSV), and an isolated **Square Sandbox Draft Shift Synchronization Engine** with unconditional production write blocks.

2. Completed Development Phases

Phase	Status	Key Capabilities & Architecture
Phase 1: Foundation & Integration	Complete	Django structure (<code>`spirit_scheduler`</code>), PostgreSQL capability, staff/role schemas, idempotent staff seed, <code>`SquareClient`</code> API wrapper, sandbox write guards, and <code>`check_secrets.py`</code> script.
Phase 2: Management Web UI & Engine	Complete	Bootstrap 5 UI, show scraper, CSV availability validator, office & 50/50 rotation seeds, deterministic constraint engine, audited overrides, local approval, and Excel/PDF/CSV exports.
Phase 3.1: Square Sandbox Sync	Complete	Pre-sync validation engine, unmapped staff/role detection, Square Sandbox conflict search, pre-sync web preview (<code>`/sync-preview/`</code>), CLI sync command (<code>`sync_schedule_to_square_sandbox`</code>), status transition to <code>`SYNCED_TO_SQUARE`</code> , and idempotency keys.

3. Technical Verification & Test Results

Test Module	Tests	Status	Focus Area
<code>`test_scheduling_engine.py`</code>	21	PASSED	Hard constraints, soft scoring, rotation rules, shortage detection
<code>`test_square_sync.py`</code>	5	PASSED	Pre-sync validation, unmapped IDs, Sandbox shift publishing, CLI command
<code>`test_square_client.py`</code>	8	PASSED	Square API requests, error mapping, search logic
<code>`test_imports_exports_workflow.py`</code>	12	PASSED	CSV availability validator, show scraper, Excel/PDF exports
<code>`test_views.py`</code>	6	PASSED	Django management views, authentication, page rendering
<code>`test_draft_shift.py`</code>	5	PASSED	Draft shift payload verification and dry-run safety
<code>`test_square_config.py`</code>	5	PASSED	Environment isolation & production blocking guards
<code>`test_models_and_seed.py`</code>	4	PASSED	Model validation & idempotent seed script execution

4. Next Roadmap Steps

- **Phase 3.2: Automated Square Timecard & Historical Hours Sync** — Import actual timecard records from Square Labor API to seed fairness scoring automatically.
- **Phase 3.3: High-Capacity Event Staffing Rules (100–175 Guests)** — Dynamic scaling rules for large show attendance.
- **Phase 3.4: Production Server Deployment Setup** — WSGI/ASGI service runners, PostgreSQL, SSL, and Mac mini deployment runbook.

Report Generated by: Antigravity AI Assistant | **Organization:** Spirit of Newfoundland Productions